

**BEZHETSK, Russian Federation**

WMO: 272170

Lat: **57.800N**Lon: **36.700E**Elev: **482**StdP: **14.44**Time Zone: **3.00 (E03)**Period: **82-93**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.8      | -13.5 | -25.6                         | 1.4 | -20.8 | -18.0 | 2.1 | -13.3 | 23.4                  | 15.6 | 21.1 | 21.4 | 4.7                   | 30   | 0.613 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.4                   | 79.5            | 67.6 | 76.4 | 65.4 | 73.6 | 63.5 | 69.3                | 77.0 | 66.9 | 74.0 | 64.9 | 71.7 | 7.4                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.3                            | 98.6 | 74.2 | 64.1 | 91.3 | 71.3 | 62.1 | 84.9 | 68.3 | 33.7          | 77.5 | 31.7 | 74.5 | 30.2 | 72.0 | 77.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |              |             |                    | n-Year Return Period Values of Extreme Temperature |              |             |              |             |              |             |              |             |     |
|-------------------|-------------|-------------|----------------------------|--------------|-------------|--------------------|----------------------------------------------------|--------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|-----|
| 1%                |             | 2.5%        | 5%                         | Mean         |             | Standard Deviation | n=5 years                                          |              | n=10 years  |              | n=20 years  |              | n=50 years  |              | DB          | WB  |
| Min               | Max         |             |                            | Min          | Max         |                    | Min                                                | Max          | Min         | Max          | Min         | Max          | Min         | Max          |             |     |
| (a)               | (b)         | (c)         |                            | (e)          | (f)         | (g)                | (h)                                                | (i)          | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          |             |     |
| (4) <b>19.2</b>   | <b>16.9</b> | <b>14.9</b> |                            | <b>-24.1</b> | <b>83.5</b> | <b>8.8</b>         | <b>2.3</b>                                         | <b>-30.4</b> | <b>85.2</b> | <b>-35.6</b> | <b>86.6</b> | <b>-40.5</b> | <b>87.9</b> | <b>-46.9</b> | <b>89.6</b> | (4) |
| (5)               |             |             |                            | <b>-22.0</b> | <b>72.6</b> | <b>10.9</b>        | <b>2.4</b>                                         | <b>-29.8</b> | <b>74.4</b> | <b>-36.2</b> | <b>75.8</b> | <b>-42.3</b> | <b>77.2</b> | <b>-50.3</b> | <b>79.0</b> | (5) |

**Monthly Climatic Design Conditions**

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 38.9   | 14.1  | 15.4  | 26.6  | 40.2  | 53.3  | 58.8  | 62.4  | 59.1  | 50.2  | 39.3  | 26.8  | 18.6  |      |
|                                                                               | DBStd                                                                         | 19.71  | 16.22 | 13.05 | 9.47  | 7.84  | 7.78  | 6.82  | 5.44  | 6.42  | 6.30  | 7.44  | 10.04 | 11.90 |      |
|                                                                               | HDD50                                                                         | 5273   | 1112  | 970   | 727   | 309   | 58    | 12    | 1     | 7     | 71    | 336   | 697   | 973   |      |
|                                                                               | HDD65                                                                         | 9616   | 1577  | 1390  | 1192  | 745   | 367   | 205   | 115   | 199   | 445   | 796   | 1147  | 1438  |      |
|                                                                               | CDD50                                                                         | 1211   | 0     | 0     | 0     | 15    | 160   | 276   | 387   | 290   | 78    | 5     | 0     | 0     |      |
|                                                                               | CDD65                                                                         | 77     | 0     | 0     | 0     | 0     | 4     | 19    | 36    | 17    | 1     | 0     | 0     | 0     |      |
|                                                                               | CDH74                                                                         | 527    | 0     | 0     | 0     | 0     | 48    | 119   | 236   | 116   | 8     | 0     | 0     | 0     |      |
|                                                                               | CDH80                                                                         | 53     | 0     | 0     | 0     | 0     | 1     | 12    | 28    | 10    | 2     | 0     | 0     | 0     |      |
| Wind                                                                          | WSAvg                                                                         | 7.1    | 8.3   | 7.9   | 7.6   | 7.5   | 6.6   | 6.2   | 5.3   | 5.4   | 6.5   | 7.0   | 8.2   | 8.2   |      |
| Precipitation                                                                 | PrecAvg                                                                       | 21.3   | 1.0   | 0.8   | 0.9   | 1.3   | 2.0   | 2.6   | 3.1   | 2.6   | 2.1   | 1.8   | 1.6   | 1.4   |      |
|                                                                               | PrecMax                                                                       | 28.3   | 2.2   | 1.7   | 2.1   | 4.0   | 3.7   | 6.7   | 7.4   | 5.0   | 5.0   | 3.4   | 3.1   | 3.4   |      |
|                                                                               | PrecMin                                                                       | 13.7   | 0.1   | 0.0   | 0.2   | 0.0   | 0.4   | 0.4   | 0.9   | 0.3   | 0.3   | 0.0   | 0.6   | 0.6   |      |
|                                                                               | PrecStd                                                                       | 3.2    | 0.5   | 0.4   | 0.4   | 0.7   | 0.9   | 1.3   | 1.5   | 1.3   | 1.0   | 0.8   | 0.7   | 0.6   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 36.7  | 40.8  | 48.4  | 67.7  | 78.9  | 81.4  | 83.1  | 81.1  | 72.9  | 60.0  | 45.3  | 38.3  |      |
|                                                                               |                                                                               | MCWB   | 35.1  | 38.3  | 43.9  | 55.6  | 64.2  | 68.1  | 69.8  | 68.3  | 61.5  | 55.2  | 42.3  | 36.7  |      |
|                                                                               | 2%                                                                            | DB     | 35.4  | 37.9  | 42.6  | 62.7  | 74.6  | 77.3  | 80.2  | 77.2  | 67.5  | 54.9  | 42.9  | 36.5  |      |
|                                                                               |                                                                               | MCWB   | 34.0  | 36.1  | 37.7  | 51.1  | 61.1  | 67.1  | 68.5  | 65.9  | 59.1  | 51.2  | 40.8  | 35.1  |      |
|                                                                               | 5%                                                                            | DB     | 34.1  | 35.3  | 39.8  | 57.7  | 70.9  | 74.2  | 76.7  | 74.0  | 64.0  | 52.0  | 40.4  | 35.1  |      |
|                                                                               |                                                                               | MCWB   | 32.8  | 34.1  | 36.2  | 48.8  | 58.8  | 64.1  | 66.7  | 64.1  | 57.1  | 49.0  | 39.0  | 33.9  |      |
|                                                                               | 10%                                                                           | DB     | 32.6  | 33.2  | 37.0  | 53.2  | 67.5  | 71.2  | 73.5  | 70.8  | 60.5  | 49.3  | 38.3  | 33.3  |      |
|                                                                               |                                                                               | MCWB   | 31.5  | 32.0  | 34.0  | 46.7  | 56.8  | 62.0  | 64.0  | 62.5  | 54.7  | 46.9  | 37.0  | 32.3  |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 36.0  | 38.8  | 44.2  | 56.1  | 65.6  | 72.1  | 72.7  | 69.8  | 63.4  | 56.8  | 42.9  | 37.2 |
|                                                                               |                                                                               |        | MCDB  | 36.4  | 40.4  | 48.2  | 67.2  | 76.3  | 81.2  | 80.5  | 79.5  | 69.3  | 58.9  | 44.4  | 38.0 |
| 2%                                                                            |                                                                               | WB     | 34.1  | 36.2  | 39.2  | 52.9  | 62.6  | 67.8  | 70.4  | 67.1  | 60.2  | 52.2  | 40.9  | 35.6  |      |
|                                                                               |                                                                               | MCDB   | 34.9  | 37.4  | 41.6  | 60.9  | 71.1  | 74.9  | 77.9  | 75.0  | 66.1  | 54.3  | 42.7  | 36.2  |      |
| 5%                                                                            |                                                                               | WB     | 32.9  | 34.2  | 36.4  | 50.0  | 60.3  | 65.6  | 67.6  | 65.0  | 57.8  | 49.6  | 39.2  | 34.1  |      |
|                                                                               |                                                                               | MCDB   | 33.8  | 35.2  | 38.7  | 56.2  | 68.7  | 72.6  | 74.6  | 72.0  | 62.8  | 51.7  | 40.4  | 34.8  |      |
| 10%                                                                           |                                                                               | WB     | 31.6  | 32.2  | 34.6  | 46.9  | 57.9  | 63.3  | 65.4  | 63.3  | 55.3  | 47.1  | 37.2  | 32.4  |      |
|                                                                               |                                                                               | MCDB   | 32.7  | 33.4  | 36.6  | 52.3  | 65.6  | 69.3  | 71.7  | 69.4  | 60.0  | 48.9  | 38.2  | 33.4  |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 11.0  | 12.0  | 12.2  | 14.1  | 17.5  | 15.1  | 15.4  | 15.6  | 13.5  | 10.2  | 8.5   | 10.4  |      |
|                                                                               |                                                                               | MCDBR  | 7.7   | 7.1   | 14.5  | 21.7  | 22.6  | 19.5  | 20.6  | 21.1  | 18.7  | 14.3  | 7.5   | 8.9   |      |
|                                                                               |                                                                               | MCWBR  | 8.0   | 6.1   | 10.3  | 12.6  | 11.3  | 10.1  | 10.3  | 11.5  | 11.2  | 10.3  | 6.6   | 8.7   |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 7.9   | 6.2   | 10.9  | 20.0  | 20.4  | 18.0  | 18.9  | 19.0  | 17.7  | 12.5  | 7.4   | 9.2   |      |
|                                                                               |                                                                               | MCWBR  | 8.2   | 5.3   | 8.3   | 12.2  | 10.8  | 10.5  | 10.3  | 11.1  | 11.3  | 9.8   | 6.6   | 9.0   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          |        | 0.278 | 0.318 | 0.336 | 0.387 | 0.376 | 0.366 | 0.402 | 0.401 | 0.363 | 0.340 | 0.309 | 0.269 |      |
|                                                                               | taud                                                                          |        | 2.091 | 2.066 | 2.121 | 2.192 | 2.358 | 2.401 | 2.320 | 2.310 | 2.395 | 2.365 | 2.214 | 2.084 |      |
|                                                                               | Ebn at Noon                                                                   |        | 174   | 218   | 253   | 259   | 272   | 277   | 262   | 253   | 246   | 216   | 163   | 139   |      |
|                                                                               | Edh at Noon                                                                   |        | 17    | 28    | 35    | 39    | 35    | 34    | 36    | 35    | 27    | 21    | 16    | 13    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        |        | 146   | 404   | 885   | 1260  | 1664  | 1734  | 1689  | 1323  | 811   | 357   | 125   | 75    |      |
|                                                                               | RadStd                                                                        |        | 25    | 72    | 105   | 135   | 131   | 141   | 208   | 113   | 106   | 65    | 20    | 19    |      |